Proshenjit Sarker

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Education

2026 – Present **PhD. in Mechanical Science and Technology,Gunma University, Japan**.

2024 – 2025 **M.Sc. in Electronics and Communication Engineering, Khulna University**.
CGPA: 3.88/4.00 (89.40% Marks), Total Credit: 40
Thesis title: *Integrated Surveillance System for Khulna University: Enhancing Safety, Security, and Sustainability through Camera Footage and Peripheral Sensors*.

2019–2023 **B.Sc. in Electronics and Communication Engineering, Khulna University**.
Four Year Honours Degree, CGPA: 3.59/4.00 (75.80% marks), Total Credits: 160.
Thesis title: *Multi-Platform Data Security Systems using n-Bit Modified (n-BM) and Reusable Block (n-RB) AES*.

Research Publications

Journal Articles

1. P. Sarker, A. Ksibi, M. M. Jamjoom, K. Choi, A. A. Nahid, and M. A. Samad, "Breast cancer prediction with feature-selected xgb classifier, optimized by metaheuristic algorithms," *Journal of Big Data*, vol. 12, no. 1, p. 78, 2025.
2. P. Sarker, K. Choi, A.-A. Nahid, and M. A. Samad, "Catboost with physics-based metaheuristics for thyroid cancer recurrence prediction," *BioData Mining*, vol. 18, no. 1, p. 84, 2025.
3. P. Sarker, A.-A. Nahid, M. A. Samad, and K. Choi, "Gradient boosting classifier with zebra optimization algorithm for pregnancy risk prediction," *ICT Express*, 2026.
4. P. Sarker, A.-A. Nahid, K. Choi, and M. A. Samad, "Feature reduction using swarm optimization and random forest classifiers for early diabetes risk prediction," *Scientific Reports*, 2026.
5. P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Metaheuristic-driven feature selection for human activity recognition on ku-har dataset using xgboost classifier," *Sensors*, vol. 25, no. 17, p. 5303, 2025.
6. P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Gallstone classification using random forest optimized by sand cat swarm optimization algorithm with shap and dice-based interpretability," *Sensors*, vol. 25, no. 17, p. 5489, 2025.
7. P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Dengue fever detection using swarm intelligence and xgboost classifier: An interpretable approach with shap and dice," *Information*, vol. 16, no. 9, p. 789, 2025.
8. P. Das, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Breast cancer prediction using rotation forest algorithm along with finding the influential causes," *Bioengineering*, vol. 12, no. 10, p. 1020, 2025.
9. P. Das, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Dengue fever classification integrating bird swarm algorithm with gradient boosting classifier along with feature selection and shap–dice based interpretability," *Applied Sciences*, vol. 15, no. 21, p. 11 413, 2025.
10. K. Shrestha, P. Sarker, J.-J. Tiang, and A.-A. Nahid, "Metaheuristic-based gallstone classification using rotational forest explained with shap," *Frontiers in Digital Health*, vol. 7, p. 1 727 559, 2025.

11 P. Das, M. A. U. Khan, P. Sarker, and A.-A. Nahid, "Gallstone disease classification using sloa-optimized catboost classifier with explainable ai," *Plos one*, vol. 21, no. 6, e0342945, 2026.

12 P. Sarker and S. S. Alam, "Cascaded yolo-based approach for bangladeshi license plate identification and number recognition," *Plos one*, vol. 21, no. 8, e0355858, 2026.

Conference Proceedings

1 M. I. H. Sikder, P. Sarker, M. T. Hasan, and M. I. Kadir, "Reusable block aes: A modified aes algorithm for the encryption and decryption of sensitive data," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2025, pp. 1–6.

2 M. I. H. Sikder, P. Sarker, and M. I. Kadir, "A smartphone application for multiformat data security using n-bit reusable block aes," in *2025 International Conference on Electrical, Computer and Communication Engineering (ECCE)*, IEEE, 2025, pp. 1–6.

Research Under Progress

1. SCSO-LGBM: A Metaheuristic-Optimized Explainable Framework for Heart Failure Survival Prediction. (**Under Review**)

Projects

App for Data Encryption — Developed an APK to securely encrypt images and text; part of my undergraduate thesis.

Height Measurement Machine — Arduino UNO–based non-contact height measurement using ultrasonic sensor and Android app.

CRC Simulator — Simulator for binary data transmission; indicates errors via syndromes.

Skills

Languages — Strong reading, writing and speaking competencies for English and Bangla.

Coding — Python (Classification; Metaheuristic Algorithms: SLO, WOA, FOX, BESA, EOA, EVOA, HBA, SCSO, and others; XAI: SHAP and LIME, and DiCE), Matlab, C/C++, JAVA, LaTeX.

Software — MS Office (Word, Excel, PowerPoint, Visio); basic Android Studio and Cisco Packet Tracer.

Embedded Systems — Arduino, NVIDIA Jetson Developer Kit.

Others — Academic research, journal writing, and LaTeX typesetting using Overleaf and TeXstudio, with several publications.

Experience

Research Assistant

2025 — Developing Security System using Jetson Developer Kit, YOLO, Viola Jones, and ViT Algorithms.

Industrial Training

2023 — Khulna 225 MW Combined Cycle Power Plant.

Experience (continued)

Certification/Award

2024 **Certificate of Appreciation**. IEEE SPICSCON 2024, for outstanding contributions as Vice Chairperson (Technical), IEEE KU SB.

2020 **Certification**. Technical Support Fundamentals, an online non-credit course authorized by Google and offered through Coursera.

2019 **Champion**. Fresher Debaters' League 2019, Khulna University.

References

Reference 01

Abdullah-Al Nahid
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Electronics and Communication Engineering Discipline, Khulna University,
Khulna, Bangladesh.
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Reference 02

Md Abdus Samad
Research Professor
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South Korea.
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